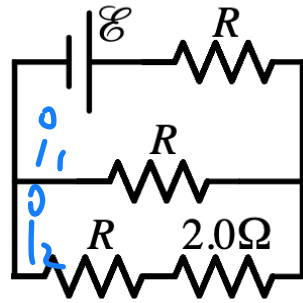


1. An ideal battery is connected to several resistors in combination as shown in the figure. The three resistors R have the same resistance. If the current through the $2.0\ \Omega$ resistor is 0.60 A , and the current through the resistor in series with the battery is 1.8 A , what is the emf of the battery?



- (1) 6.0 V
- (2) 3.6 V
- (3) 0.4 V
- (4) 2.0 V
- (5) 1.2 V

The two bottom rows are in parallel $\Rightarrow$ the voltage through them is equal.

$$V = i_1 R = i_2 (R + 2\ \Omega) = 0.6\text{ A} (R + 2\ \Omega)$$

We also know: $i_1 + 0.6\text{ A} = 1.8\text{ A} \Rightarrow i_1 = 1.2\text{ A}$

This means that $1.2\text{ A} R = 0.6\text{ A} (R + 2\ \Omega) \Rightarrow R = 2\ \Omega$

Resistance of bottom row: $R + 2\ \Omega = 4\ \Omega$

Resistance for bottom two rows: $\frac{1}{\frac{1}{R} + \frac{1}{4\ \Omega}} = \frac{4}{3}\ \Omega$

Total resistance: $R_{\text{tot}} = R + \frac{4}{3}\ \Omega = \frac{10}{3}\ \Omega$

Then total EMF: $\mathcal{E} = 1.8\text{ A} \cdot \frac{10}{3}\ \Omega = 6\text{ V}$

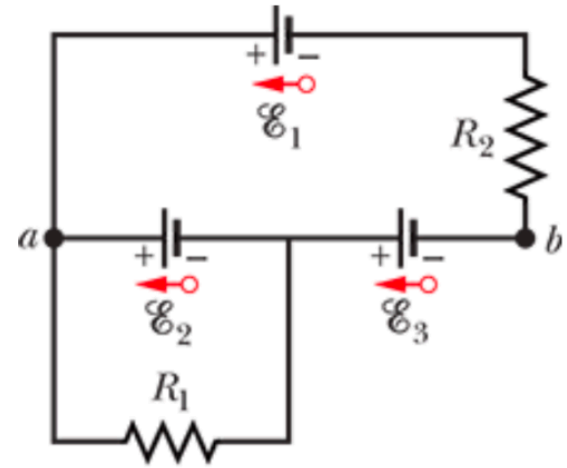
2. In the figure $\mathcal{E}_1 = 2.0$ V, while $\mathcal{E}_2 = \mathcal{E}_3 = 3$ V. The resistances are $R_1 = 1$ k Ω and $R_2 = 2$ k Ω . What is the magnitude of the current flowing through resistor 2, in mA?

- (1) 2
- (2) 3.5
- (3) 0.5
- (4) 11
- (5) 30

Looking at the top loop:

$$\mathcal{E}_1 - \mathcal{E}_2 - \mathcal{E}_3 - i R_2 = 0$$

$$i = \frac{-4\text{V}}{2\text{k}\Omega} = -2 \times 10^{-3} \text{ A} = -2 \text{ mA}$$



3. A capacitor with an initial potential difference of 50 V is discharged through a resistor when a switch between them is closed at $t = 0$. At $t = 1.0$ s, the potential difference across the capacitor is 5 V. What is the potential difference, in mV, across the capacitor at $t = 2.0$ s?

- (1) 500 (2) 650 (3) 150 (4) 30 (5) 0

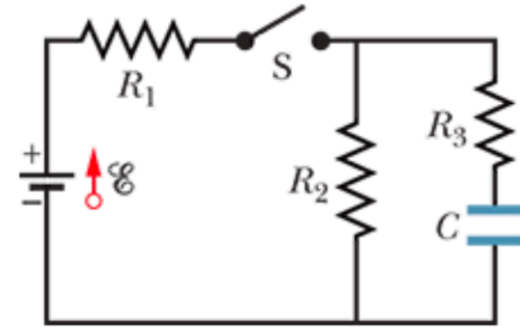
The change is exponential $\Rightarrow$ over a given time it is reduced by a given factor. If it is reduced by

$\frac{50V}{5V}$ = a factor of 10 in the first second, it should have

the same reduction in the second second:

$$\frac{5V}{\Delta V} = \frac{50V}{5V} \Rightarrow \Delta V = 0.5V = 500 \text{ mV}$$

4. In the figure, $\mathcal{E} = 3.0 \text{ V}$, $C = 500 \mu\text{F}$, $R_1 = 1.0 \text{ k}\Omega$, $R_2 = 2.0 \text{ k}\Omega$ and $R_3 = 5.0 \text{ k}\Omega$. At $t = 0$ the switch is closed. After a long enough time for the capacitor to become fully charged, what will be its final charge, in mC?



- (1) 1
- (2) 7
- (3) 11
- (4) 20
- (5) 130

After a long time, there will be no current through the capacitor, and the potential difference through $R_3 + C$ will be the same as through R_2 since they are in parallel

Current in left loop: $i = \frac{3\text{V}}{1\text{k}\Omega + 2\text{k}\Omega} = 1\text{mA}$

$\Rightarrow$ potential diff. through R_2 : $\Delta V = 1\text{mA} \cdot 2\text{k}\Omega = 2\text{V}$

Since there is no current through R_3 all ΔV will fall on capacitor

$$\Delta V = 2\text{V} = \frac{Q}{C} \Rightarrow Q = 2\text{V} \cdot 500 \mu\text{F} = 1\text{mC}$$

5. A horizontal power line carries a current of 5.0 kA from east to west. Earth's magnetic field of magnitude $60 \mu\text{T}$ is directed toward the north and is inclined downward from horizontal at an angle of 30° . What is the magnitude of the magnetic force per unit length on the power line, in N/m?

- (1) 0.3 (2) 0.01 (3) 0.1 (4) 0.03 (5) 0.6

Power line and magnetic field are perpendicular.

$$F = I \cdot \vec{L} \times \vec{B} = I L B$$

$$\Rightarrow \frac{F}{L} = IB = 5 \text{ kA} \cdot 60 \mu\text{T} = 0.3 \text{ N/m}$$

6. In the above problem, what is the direction of the force?

(1) none of the others

(2) east

(3) west

(4) up

(5) down

Since the magnetic field is inclined 30° downwards, the force will be 60° from horizontal $\Rightarrow$ will not align with any of the directions listed.

7. An electron moves through a uniform magnetic field given by $\vec{B} = B_x \hat{i} + (2B_x) \hat{j}$. At a particular instant, the electron has velocity $\vec{v} = (2\hat{i} + 3\hat{j})$ m/s and the magnetic force acting on it is $(4.8 \times 10^{-19}) \hat{k}$ N. Find B_x in T.

(1) 3.0

(2) 1.0

(3) 4.5

(4) 5.5

(5) 15.0

$$\vec{F} = q \vec{v} \times \vec{B} = q_e (2 \cdot 2B_x - 3B_x) \hat{k} = q_e B_x \hat{k}$$

$$\Rightarrow B_x = \frac{4.8 \times 10^{-19} \text{ N}}{1.6 \times 10^{-19} \text{ C}} = 3 \text{ T}$$

8. A square-shaped loop with sides of 1 m, carrying a current of 2 A, is placed in a uniform magnetic field of 0.1 T. What is the work required to flip the loop from its lowest-energy orientation to its highest-energy orientation, in J?

- (1) 0.4 (2) 1.0 (3) 0.2 (4) 0.1 (5) 2.2

$$\mu = iA = 2 \text{ Am}^2$$

$$U_{\max} = \mu B$$

$$U_{\min} = -\mu B$$

$$\Rightarrow \text{total energy needed: } 2\mu B = 2 \cdot 2 \text{ Am}^2 \cdot 0.1 \text{ T} = 0.4 \text{ J}$$

9. An electron is traveling in the $+y$ direction at 25 m/s . It enters a region of space in which there is a uniform magnetic field of 0.2 T is pointing in the $-x$ direction. To keep the electron moving on a straight line without deflection, how large an electric field (in N/C) must be applied and what is its direction?

- (1) 5, into the page (2) 10, $+x$ direction (3) 5, out of the page (4) 4, $-x$ direction (5) 1.8, $-y$ direction

Magnetic force: $\vec{F}_B = q_e \cdot 25 \text{ m/s} \cdot 0.2 \text{ T} = q_e \cdot 5 \text{ N/C}$

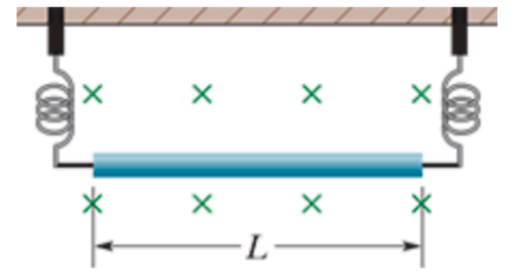
Electric force: $\vec{F}_E = q_e \cdot \vec{E}$

$\Rightarrow$ They are equal if $E = 5 \text{ N/C}$

magnetic force: $\vec{v} \times \vec{B}$ points out of the page

$\Rightarrow E$ needs to point into the page

10. A 25 g wire of length $L = 41$ cm is suspended horizontally by a pair of flexible leads in a uniform magnetic field of magnitude 100 mT, as shown in the figure. What are the magnitude (in A) and direction (left or right) of the current required to remove the tension in the supporting leads?



(1) 6, right

(2) 10, left

(3) 2, right

(4) 8, left

(5) 15, right

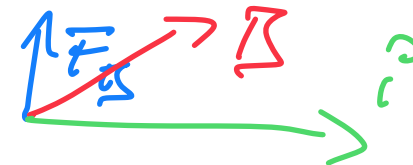
$$\vec{F}_B = i L \vec{B}$$

$$F_g = m \cdot g$$

we want $\vec{F}_B = \vec{F}_g$

$$\Rightarrow i = \frac{mg}{L B} = \frac{0.025 \cdot 9.81}{0.41 \cdot 0.1} = 5.98 \text{ A}$$

We want $\vec{F}_B$ to point up



11. A long solenoid with 53 turns/cm and a radius of 9 cm carries a current of 30 mA. A current of 2 A exists in a straight conductor located along the central axis of the solenoid. At what radial distance, in mm, from the axis will the direction of the resulting magnetic field be at 45° to the axial direction?

(1) 3.0

(2) 2.0

(3) 0.5

(4) 10.0

(5) 15.0

B_{solenoid} points along the solenoid, while B_{wire} will be perpendicular.
 $\rightarrow 45^\circ$ means that their magnitude should be equal.

$$B_{\text{solenoid}} = \mu_0 n i_s \Rightarrow \mu_0 n i_s = \frac{\mu_0 i_w}{2\pi R} \Rightarrow R = \frac{i_w}{2\pi n i_s} =$$
$$B_{\text{wire}} = \frac{\mu_0 i}{2\pi R} = \frac{2}{2\pi \cdot 5300 \cdot 0.03} = 0.002 \text{ m}$$

12. A long cylindrical conductor of radius $a = 2$ cm carries a current of 2 A uniformly distributed across the conductor's cross-section. At what distance from the center of the wire is the magnitude of the magnetic field one third of the magnetic field at the surface of the wire?

(1) $R/3$ and $3R$

(2) $R/\sqrt{3}$ and $\sqrt{3}R$

(3) only $3R$

(4) only $R/3$

(5) only $\sqrt{3}R$

$$\mu_0 \cdot i_{\text{enclosed}} = 2\pi r B \quad (\text{since this is axially symmetric})$$

For r distance from center:

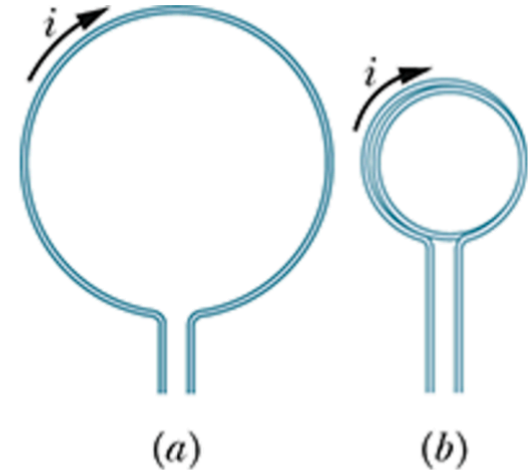
$$\boxed{r < a} : i_{\text{enclosed}} = i_{\text{total}} \cdot \frac{r^2}{a^2} \Rightarrow \frac{B(r)}{B(a)} = \frac{\frac{i_{\text{enclosed}}}{r}}{\frac{i_{\text{total}}}{a}} = \frac{r}{a}$$

$\Rightarrow$ we will get a third of magnetic field at $R/3$

$$\boxed{r > a} : i_{\text{enclosed}} = i_{\text{total}} \Rightarrow \frac{B(r)}{B(a)} = \frac{1/r}{1/a} = \frac{a}{r}$$

$\Rightarrow$ we will get $1/3 B$ at $3R$ distance

13. Figure (a) shows a length of wire carrying a current i and bent into a circular coil of one turn. In Figure (b) the same length of wire that carries the same current i has been bent to give a coil of two turns, each of half the original radius. If B_a and B_b are the magnitudes of the magnetic fields at the centers of the two coils, what is the ratio B_b/B_a ?



(1) $1/4$

(2) 4

(3) 1

(4) $1/2$

(5) 2

$$B_a = \frac{\mu_0 i}{2R}$$

$$B_b = \frac{\mu_0 2i}{2(R/2)} = 4B_a$$

$$\Rightarrow \frac{B_b}{B_a} = 4$$

14. A rectangular loop of wire (3cm - 4cm) is placed inside a solenoid of radius 8 cm. The solenoid is 25 cm long and is wound with 1500 turns of wire carrying 4 A of current. The normal to the loop lies along the solenoid axis. If the current in the solenoid is reduced to zero in 0.2 sec., find the magnitude of the average induced emf in the loop.

(1) 0.18 mV

(2) 36 mV

(3) 1.8 mV

(4) 0.25 mV

(5) 80 mV

initial magnetic field in solenoid: $B = \mu_0 \frac{N}{L} i = 4\pi \times 10^{-7} \cdot \frac{1500}{0.25} \cdot 4 = 0.03 \text{ T}$

magnetic flux through loop:

$$\Phi = 0.03 \text{ T} \cdot 0.03 \text{ m} \cdot 0.04 \text{ m} = 3.6 \times 10^{-5} \text{ Tm}^2$$

$$\text{EMF} = -\frac{\Delta \Phi}{\Delta t} = \frac{3.6 \times 10^{-5} \text{ Tm}^2}{0.2 \text{ s}} = 0.00018 \text{ V} = 0.18 \text{ mV}$$

15. Consider an serial RLC circuit with $R = 10\ \Omega$, $L = 1.0\ \text{H}$, and $C = 1.0\ \mu\text{F}$. It is connected to a voltage source which produces $V_0 = 10\ \text{V} \sin \omega_0 t$ where ω_0 is the angular resonance frequency of the circuit. Find the amplitude of the voltage across the inductor.

(1) 1000 V

(2) 2 V

(3) 70 V

(4) 10 V

(5) 400 V

ω_0 is the resonance frequency $\Rightarrow \omega_0 = \frac{1}{\sqrt{LC}} = 1000\ \text{rad/s}$

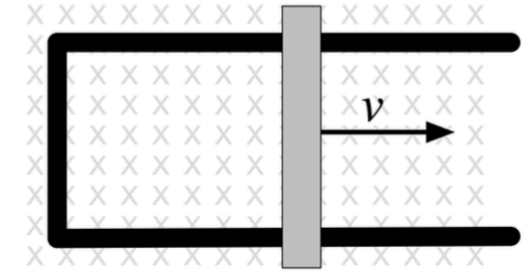
At resonance, the current amplitude is

$$I_0 = \frac{V_0}{R} = 1\ \text{A}$$

The voltage amplitude across the inductor is

$$V_L = I_0 \cdot X_L = I_0 \omega_0 L = 1 \cdot 1000 \cdot 1 = 1000\ \text{V}$$

16. A rod of length L with resistance R lies across frictionless conducting rails in a constant uniform magnetic field B , as shown. Assume the rails have negligible resistance. The rod is pulled to the right by a person with constant power P . After a long time, what is the resulting speed of the rod?



(1) $\sqrt{RP}/BL$

(2) P/RBL

(3) P/RB^2L^2

(4) R/PBL

(5) $\sqrt{BP}/RL$

Dissipation of energy per unit time in circuit: $P = \frac{V^2}{R}$

This power is supplied by the person.

The voltage in the circuit is: $V = -\frac{d\Phi}{dt} = B \cdot L \cdot v$

$$\Rightarrow P = \frac{(BLv)^2}{R} \Rightarrow v = \frac{\sqrt{RP}}{BL}$$

17. A $15\mu\text{F}$ capacitor in an oscillating LC circuit stores a maximum energy of $20\mu\text{J}$. If the resonance frequency of the circuit is $f_0 = 3\text{kHz}$, what is the maximum current in mA through the inductor?

(1) 462

(2) 624

(3) 327

(4) 57

(5) 212

$$\text{Energy in inductor} = \frac{1}{2} L i^2$$

$$\text{Resonance frequency: } \omega_0 = 2\pi f = \frac{1}{\sqrt{LC}} \Rightarrow L = \frac{1}{15 \times 10^{-6} \cdot (2\pi \cdot 3000)^2} = 1.87 \times 10^{-4} \text{ H}$$

At max current all energy is stored in inductor:

$$20\mu\text{J} = \frac{1}{2} L \cdot i_{\text{max}}^2$$
$$\Rightarrow i_{\text{max}} = \sqrt{\frac{2 \cdot 20\mu\text{J}}{1.87 \times 10^{-4} \text{ H}}} = 0.462 \text{ A}$$

18. A $1.50 \mu\text{F}$ capacitor is connected to an AC generator with $\mathcal{E} = 30 \text{ V}$. What is the amplitude of the resulting alternating current in mA if the frequency of the EMF is 1 kHz?

- (1) 280 (2) 110 (3) 45 (4) 8.0 (5) 2.0

$$X_C = \frac{1}{\omega C} = \frac{1}{2\pi f C} = \frac{1}{2\pi \cdot 1000 \text{ Hz} \cdot 1.5 \times 10^{-6} \text{ F}} = 106 \frac{\Omega}{\text{F}}$$

$$i_{\text{max}} = \frac{\mathcal{E}}{X_C} = \frac{30}{106} = 0.28 \text{ A} = 280 \text{ mA}$$

19. An RLC-circuit ($R = 50 \, \Omega$, $L = 100 \, \text{mH}$, $C = 20 \, \mu\text{F}$,) is connected to an alternating emf $\mathcal{E} = (100 \, \text{V})\sin(\omega_d t)$, where $\omega_d = 1000 \, \text{s}^{-1}$. What is amplitude of the current oscillating in the circuit?

(1) 1.4 A

(2) 6.0 A

(3) 3.3 A

(4) 2.1 A

(5) 4.2 A

$$Z = \sqrt{R^2 + (X_L - X_C)^2} = \sqrt{R^2 + \left(\omega_d L - \frac{1}{\omega_d C}\right)^2} = \\ = \sqrt{50^2 + \left(1000 \cdot 0.1 - \frac{1}{1000 \cdot 20 \cdot 10^{-6}}\right)^2} = 70.7 \, \Omega$$

$$I_{\text{max}} = \frac{\mathcal{E}_{\text{max}}}{Z} = \frac{100}{70.7} = 1.41 \, \text{A}$$

20. A transformer has 500 primary turns and 10 secondary turns. If the primary voltage is $V_p = 120 \text{ V}$ (rms), what is the secondary voltage V_s in V with an open circuit?

(1) 2.4

(2) 3.6

(3) 4.0

(4) 1.2

(5) 0.6

$$V_s = V_p \cdot \frac{N_s}{N_p} = 120 \cdot \frac{10}{500} = 2.4 \text{ V}$$